

Solutions to UGB Mock 2

Q.1

Problem: Let x, y, z be positive integers such that $\gcd(x, y, z) = 1$, and the expression

$$\frac{xy}{z} + \frac{yz}{x} + \frac{zx}{y}$$

is an integer. Prove that xyz is a perfect square.

Solution: Let the given expression be denoted by N . Thus, $N = \frac{x^2y^2 + y^2z^2 + z^2x^2}{xyz} \in \mathbb{Z}$. This implies that xyz perfectly divides $x^2y^2 + y^2z^2 + z^2x^2$. We will analyze this divisibility using p -adic valuations. Let p be an arbitrary prime number, and let $v_p(n)$ denote the exponent of the highest power of p that divides n . Since N is an integer, we must have:

$$v_p(xyz) \leq v_p(x^2y^2 + y^2z^2 + z^2x^2)$$

Because $\gcd(x, y, z) = 1$, there is no prime that divides all three integers. Therefore, for our chosen prime p , at least one of the variables is not divisible by p . Without loss of generality, let us assume $p \nmid z$, meaning $v_p(z) = 0$. Let $v_p(x) = \alpha$ and $v_p(y) = \beta$. The valuation of the denominator is:

$$v_p(xyz) = \alpha + \beta + 0 = \alpha + \beta$$

The terms in the numerator sum have the following valuations:

1. $v_p(x^2y^2) = 2\alpha + 2\beta$
2. $v_p(y^2z^2) = 2\beta$
3. $v_p(z^2x^2) = 2\alpha$

By the properties of p -adic valuations, the valuation of a sum $A + B + C$ is equal to the strict minimum of their individual valuations unless the minimum is achieved by more than one term. If $\alpha \neq \beta$, then the minimum of $\{2\alpha + 2\beta, 2\beta, 2\alpha\}$ is strictly either 2α or 2β . Suppose $\alpha < \beta$. Then the minimum is 2α , leading to $v_p(\text{numerator}) = 2\alpha$. The divisibility condition requires $2\alpha \geq \alpha + \beta$, which implies $\alpha \geq \beta$, a contradiction. By symmetry, assuming $\beta < \alpha$ leads to a similar contradiction ($\beta \geq \alpha$).

Therefore, it must be true that $\alpha = \beta$. Since $\alpha = \beta$, the valuation of the product xyz for the prime p is $v_p(xyz) = \alpha + \alpha + 0 = 2\alpha$, which is an even integer. Because this holds true for *every* prime divisor p , all prime factors of xyz occur with even exponents. Thus, xyz is a perfect square. ■

Q. 2

Problem: Prove that for any positive integer $n > 1$, the equation

$$1 + 2x + 3x^2 + \cdots + nx^{n-1} = n^2$$

has a rational root between 1 and 2.

Solution: Let $f(x) = 1 + 2x + 3x^2 + \cdots + nx^{n-1}$. We recognize this series as the derivative of a finite geometric series. For $x \neq 1$:

$$\sum_{k=0}^n x^k = \frac{x^{n+1} - 1}{x - 1}$$

Differentiating both sides with respect to x yields:

$$f(x) = \frac{d}{dx} \left(\frac{x^{n+1} - 1}{x - 1} \right) = \frac{(n+1)x^n(x-1) - (x^{n+1} - 1)}{(x-1)^2} = \frac{nx^{n+1} - (n+1)x^n + 1}{(x-1)^2}$$

We claim that the rational root takes the form $x = \frac{n+1}{n}$. To establish this claim, let us evaluate $f\left(\frac{n+1}{n}\right)$. The numerator, i.e. $nx^{n+1} - (n+1)x^n + 1$, when we put $x = \frac{n+1}{n}$, becomes:

$$n \left(\frac{n+1}{n} \right)^{n+1} - (n+1) \left(\frac{n+1}{n} \right)^n + 1 = \frac{(n+1)^{n+1}}{n^n} - \frac{(n+1)^{n+1}}{n^n} + 1 = 1$$

Similarly, the denominator $((x-1)^2)$ becomes:

$$\left(\frac{n+1}{n} - 1 \right)^2 = \left(\frac{1}{n} \right)^2 = \frac{1}{n^2}$$

Dividing the numerator by the denominator, we get $f\left(\frac{n+1}{n}\right) = \frac{1}{1/n^2} = n^2$. Thus, $x = \frac{n+1}{n}$ is an exact rational root of the given equation.

Finally, we must verify that this root lies between 1 and 2. Since $n > 1$ is a positive integer, we have:

$$x = \frac{n+1}{n} = 1 + \frac{1}{n}$$

Since $n \geq 2$, the term $\frac{1}{n}$ is strictly positive and ≤ 0.5 . Thus, $1 < 1 + \frac{1}{n} \leq 1.5 < 2$. The root is indeed rational and strictly between 1 and 2. ■

Q. 3

Problem: Let $f : [0, 1] \rightarrow \mathbb{R}$ be a continuous function such that $\int_0^1 f(x)dx = \int_0^1 xf(x)dx =$

1. Prove that $\int_0^1 (f(x))^2 dx \geq 4$.

Solution: We will use the property that the integral of a squared real function is non-negative. We construct a quadratic inequality by projecting $f(x)$ onto a linear polynomial $ax + b$. For any real constants a and b :

$$\int_0^1 (f(x) - (ax + b))^2 dx \geq 0$$

Expanding this integral gives:

$$\int_0^1 (f(x))^2 dx - 2 \int_0^1 f(x)(ax + b) dx + \int_0^1 (ax + b)^2 dx \geq 0$$

Using the linearity of integrals and our given conditions ($\int_0^1 f(x) dx = 1$ and $\int_0^1 xf(x) dx = 1$),

$$\int_0^1 f(x)(ax + b) dx = a \int_0^1 xf(x) dx + b \int_0^1 f(x) dx = a(1) + b(1) = a + b$$

Now we compute the integral of the squared linear term:

$$\int_0^1 (ax + b)^2 dx = \int_0^1 (a^2x^2 + 2abx + b^2) dx = \frac{a^2}{3} + ab + b^2$$

Substituting these back into our inequality:

$$\int_0^1 (f(x))^2 dx - 2(a + b) + \left(\frac{a^2}{3} + ab + b^2 \right) \geq 0$$

To find the tightest bound, we optimize the choice of a and b . We want to maximize $2(a + b) - \left(\frac{a^2}{3} + ab + b^2 \right)$. Taking partial derivatives with respect to a and b and setting them to zero gives the system:

$$\begin{aligned} 2 - \frac{2a}{3} - b &= 0 \implies \frac{2a}{3} + b = 2 \\ 2 - a - 2b &= 0 \implies a + 2b = 2 \end{aligned}$$

Solving this system yields $a = 6$ and $b = -2$. Substituting these optimal values back into the expression:

$$\begin{aligned} \int_0^1 (f(x))^2 dx &\geq 2(6 - 2) - \left(\frac{36}{3} + (6)(-2) + (-2)^2 \right) \\ \int_0^1 (f(x))^2 dx &\geq 8 - (12 - 12 + 4) = 8 - 4 = 4 \end{aligned}$$

Thus, $\int_0^1 (f(x))^2 dx \geq 4$. ■

Q. 4

Problem: Consider three distinct points A, B, C on the rectangular hyperbola $xy = 1$. Prove that the orthocenter of $\triangle ABC$ also lies on the hyperbola.

Solution: Let the coordinates of the three distinct points on the hyperbola $xy = 1$ be parametrically defined as $A = (a, \frac{1}{a})$, $B = (b, \frac{1}{b})$, and $C = (c, \frac{1}{c})$.

First, we find the slope of the line segment BC :

$$m_{BC} = \frac{\frac{1}{c} - \frac{1}{b}}{c - b} = \frac{\frac{b-c}{bc}}{c-b} = -\frac{1}{bc}$$

The altitude from A to BC is perpendicular to BC , so its slope is the negative reciprocal, $m_A = bc$. The equation of the altitude from A is:

$$y - \frac{1}{a} = bc(x - a) \implies y = bcx - abc + \frac{1}{a}$$

Similarly, by symmetry, the equation of the altitude from B to AC is:

$$y = acx - abc + \frac{1}{b}$$

The orthocenter $H(x_H, y_H)$ is the intersection of these two altitudes. Equating the two expressions for y :

$$bcx_H - abc + \frac{1}{a} = acx_H - abc + \frac{1}{b}$$

$$bcx_H - acx_H = \frac{1}{b} - \frac{1}{a}$$

$$cx_H(b - a) = \frac{a - b}{ab}$$

Since A and B are distinct, $a \neq b$, so we can divide by $(b - a)$:

$$cx_H = -\frac{1}{ab} \implies x_H = -\frac{1}{abc}$$

Now, substitute x_H back into the altitude equation to find y_H :

$$y_H = bc \left(-\frac{1}{abc} \right) - abc + \frac{1}{a} = -\frac{1}{a} - abc + \frac{1}{a} = -abc$$

The coordinates of the orthocenter are $H \left(-\frac{1}{abc}, -abc \right)$. Checking if it lies on the hyperbola $xy = 1$, we evaluate the product of its coordinates:

$$x_H \cdot y_H = \left(-\frac{1}{abc} \right) (-abc) = 1$$

Since the product is exactly 1, the orthocenter H strictly lies on the rectangular hyperbola. ■

Q. 5

Problem: Let n be any natural number. Define $S_n := \frac{1}{n+1} \sum_{k=0}^n \frac{1}{\binom{n}{k}}$.

(a) Prove that $2S_{n+1} - S_n = \frac{2}{n+2}$, for all $n \in \mathbb{N}$.

(b) Using part (a) or otherwise, prove that $S_n = \frac{1}{2^{n+1}} \sum_{k=1}^{n+1} \frac{2^k}{k}$ for all $n \in \mathbb{N}$.

Proof of (a): We utilize the known combinatorial identity: $\frac{1}{\binom{n}{k}} = \frac{n+1}{n+2} \left(\frac{1}{\binom{n+1}{k}} + \frac{1}{\binom{n+1}{k+1}} \right)$. Substituting this into the definition of S_n :

$$S_n = \frac{1}{n+1} \sum_{k=0}^n \frac{n+1}{n+2} \left(\frac{1}{\binom{n+1}{k}} + \frac{1}{\binom{n+1}{k+1}} \right) = \frac{1}{n+2} \left(\sum_{k=0}^n \frac{1}{\binom{n+1}{k}} + \sum_{k=0}^n \frac{1}{\binom{n+1}{k+1}} \right)$$

Notice that these sums correspond almost entirely to S_{n+1} , just missing boundary terms.

$$\sum_{k=0}^n \frac{1}{\binom{n+1}{k}} = (n+2)S_{n+1} - \frac{1}{\binom{n+1}{n+1}} = (n+2)S_{n+1} - 1.$$

And,

$$\sum_{k=0}^n \frac{1}{\binom{n+1}{k+1}} = \sum_{j=1}^{n+1} \frac{1}{\binom{n+1}{j}} = (n+2)S_{n+1} - \frac{1}{\binom{n+1}{0}} = (n+2)S_{n+1} - 1.$$

Adding these together:

$$S_n = \frac{1}{n+2} \left(2(n+2)S_{n+1} - 2 \right) = 2S_{n+1} - \frac{2}{n+2}$$

Rearranging gives the desired result: $2S_{n+1} - S_n = \frac{2}{n+2}$. ■

Proof of (b): From part (a), we have the recurrence $S_{n+1} = \frac{1}{2}S_n + \frac{1}{n+2}$. Multiplying the entire equation by 2^{n+1} creates a telescoping form:

$$2^{n+1}S_{n+1} = 2^nS_n + \frac{2^{n+1}}{n+2}$$

Let $T_n = 2^nS_n$. The recurrence is now $T_{n+1} - T_n = \frac{2^{n+1}}{n+2}$. We can express T_n by summing the differences from an index $m = 1$ up to n :

$$T_n - T_0 = \sum_{m=0}^{n-1} (T_{m+1} - T_m) = \sum_{m=0}^{n-1} \frac{2^{m+1}}{m+2}$$

Letting $k = m + 2$, we rewrite the sum:

$$T_n - T_0 = \sum_{k=2}^{n+1} \frac{2^{k-1}}{k} = \frac{1}{2} \sum_{k=2}^{n+1} \frac{2^k}{k}$$

Evaluating the base case T_0 : $S_0 = \frac{1}{0+1} \cdot \frac{1}{\binom{0}{0}} = 1$, so $T_0 = 2^0 S_0 = 1$. Thus, $T_n = 1 + \frac{1}{2} \sum_{k=2}^{n+1} \frac{2^k}{k} = \frac{1}{2}(2) + \frac{1}{2} \sum_{k=2}^{n+1} \frac{2^k}{k} = \frac{1}{2} \sum_{k=1}^{n+1} \frac{2^k}{k}$. Substitute back $T_n = 2^n S_n$ and divide by 2^n :

$$S_n = \frac{1}{2^{n+1}} \sum_{k=1}^{n+1} \frac{2^k}{k}$$

Which completes the proof. ■

Q. 6

Problem: Let $f : (0, \infty) \rightarrow \mathbb{R}$ be a differentiable function such that $\lim_{x \rightarrow \infty} f'(x) = 0$. Prove that $\lim_{x \rightarrow \infty} (f(x+1) - f(x)) = 0$.

Solution: By the Mean Value Theorem, for any real number $x > 0$, the differentiable function f on the closed interval $[x, x+1]$ must have at least one point $c_x \in (x, x+1)$ such that:

$$f(x+1) - f(x) = f'(c_x) \cdot ((x+1) - x) = f'(c_x).$$

We evaluate the limit of both sides as x approaches infinity. Because c_x is strictly bounded by $x < c_x < x+1$, the Squeeze Theorem dictates that as $x \rightarrow \infty$, the value of $c_x \rightarrow \infty$ as well. Therefore:

$$\lim_{x \rightarrow \infty} (f(x+1) - f(x)) = \lim_{x \rightarrow \infty} f'(c_x) = \lim_{c_x \rightarrow \infty} f'(c_x)$$

We are explicitly given that $\lim_{t \rightarrow \infty} f'(t) = 0$. Substituting this into our limit evaluates it perfectly to 0. ■

Q. 7

Problem: On the quadratic polynomial $P(x) = ax^2 + bx + c$, you are allowed to make two types of transformations:

- (a) You can swap a and c .
- (b) You can replace x with $x + t$, where t can be any real number.

Prove that, applying these transformations repeatedly (in any order, any number of times), you can never convert the polynomial $x^2 + x + 1$ to the polynomial $x^2 + 1$.

Solution: We solve this by defining an algebraic invariant. Consider the discriminant of a quadratic polynomial $ax^2 + bx + c$, defined as $\Delta = b^2 - 4ac$. Let us observe how Δ changes under both operations.

- **Operation A (Swap a and c):** The new polynomial is $cx^2 + bx + a$. Its discriminant is $b^2 - 4(c)(a) = b^2 - 4ac$. The discriminant is entirely unchanged.
- **Operation B (Shift $x \mapsto x + t$):** The new polynomial is $a(x + t)^2 + b(x + t) + c = ax^2 + (2at + b)x + (at^2 + bt + c)$. The new discriminant is:

$$\Delta_{\text{new}} = (2at + b)^2 - 4a(at^2 + bt + c) = 4a^2t^2 + 4abt + b^2 - 4a^2t^2 - 4abt - 4ac = b^2 - 4ac$$

Again, the discriminant is strictly conserved.

Because neither allowable move alters the discriminant, Δ is invariant across any sequence of transformations. For the initial polynomial $P_{\text{start}}(x) = x^2 + x + 1$, the discriminant is $\Delta_{\text{start}} = 1^2 - 4(1)(1) = -3$. For the target polynomial $P_{\text{end}}(x) = x^2 + 1$, the discriminant is $\Delta_{\text{end}} = 0^2 - 4(1)(1) = -4$. Since $-3 \neq -4$, it is structurally impossible to reach the target polynomial. ■

Q. 8

Problem: Suppose you are given two consecutive perfect cubes whose difference is a perfect square, say k^2 . Then show that k must be the sum of two consecutive perfect squares.

Solution: Let the two consecutive perfect cubes be $(n + 1)^3$ and n^3 . We are given:

$$(n + 1)^3 - n^3 = k^2 \implies 3n^2 + 3n + 1 = k^2$$

To simplify, multiply the entire equation by 4:

$$\begin{aligned} 12n^2 + 12n + 4 &= 4k^2 \\ \implies 3(4n^2 + 4n + 1) + 1 &= 4k^2 \\ \implies 3(2n + 1)^2 + 1 &= 4k^2 \implies 4k^2 - 1 = 3(2n + 1)^2 \end{aligned}$$

Factoring the difference of squares on the left side gives:

$$(2k - 1)(2k + 1) = 3(2n + 1)^2$$

Notice that $(2k - 1)$ and $(2k + 1)$ are consecutive odd integers. Any common divisor must divide their difference, which is 2. Since both are odd, their greatest common divisor is strictly $\gcd(2k - 1, 2k + 1) = 1$. Their product equals 3 times a perfect square, $(2n + 1)^2$.

By the Fundamental Theorem of Arithmetic (Unique Factorization), since the factors are coprime, the prime factor 3 must belong entirely to one factor, and the remaining parts of both factors must be perfect squares. This leaves two cases:

- **Case 1:** $2k - 1 = 3a^2$ and $2k + 1 = b^2$

If $2k + 1 = b^2$, we can analyze this modulo 3. Since $2k - 1 = 3a^2 \equiv 0 \pmod{3}$, we have $2k + 1 = (2k - 1) + 2 \equiv 2 \pmod{3}$. However, the perfect square b^2 must be congruent to either 0 or 1 $\pmod{3}$. A perfect square cannot be 2 $\pmod{3}$. This case yields a contradiction and is impossible.

- **Case 2:** $2k - 1 = a^2$ and $2k + 1 = 3b^2$

Here, $2k - 1 = a^2$. Since $2k - 1$ is an odd number, a^2 is odd, which means a must be an odd integer. We can write $a = 2m + 1$ for some integer m . Substituting this gives:

$$\begin{aligned} 2k - 1 &= (2m + 1)^2 = 4m^2 + 4m + 1 \\ \Rightarrow 2k &= 4m^2 + 4m + 2 \end{aligned}$$

Dividing by 2, we obtain:

$$k = 2m^2 + 2m + 1 = m^2 + (m^2 + 2m + 1) = m^2 + (m + 1)^2$$

Thus, k is perfectly represented as the sum of two consecutive perfect squares, m^2 and $(m + 1)^2$. ■
